KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2024
Department of Computer Science and Engineering
CSE 4127
Image Processing and Computer Vision

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Briefly describe the role of different components of an image processing system. (06) [CO1]
   b) A lunar rover captured a 4-bit image of moon-surface under poor lighting condition, then it transmitted multiple 6×6 RLC patches. You are given an RLC image of a patch and a PDF. Your task is to adjust the pixel intensity according to the given PDF, then stretch the contrast of the output/obtained image to its maximum possible range. (20) [CO3]
   RLC image: <3,3>, <4,2>, <3,3>, <6,4>, <5,4>,
   <3,5>, <7,2>, <4,5>, <6,4>, <7,4>
   PDF:

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	0.06	0.07	0.1	0.15	0.2	0.15	0.1	0.07	0.05	0.05

   c) Prove that 'convolution in time domain corresponds to multiplication in frequency domain'. (09) [CO2]

2. a) Explain the impact of wrong choice of order for contra-harmonic mean filter. (05) [CO2]
   b) All the MRI images captured by an MRI machine contains periodic noises of different frequencies within the band (6 to 10). Design a butterworth filter to visualize the noise pattern. (10) [CO4]
   c) Given a grayscale 4×4 image of $f(x,y)$ and a quantization matrix Q. Apply JPEG algorithm and compute the RMS error of compression-decompression. (20) [CO3]

$$f(x,y) = \begin{bmatrix} 52 & 55 & 61 & 66 \\ 70 & 61 & 64 & 73 \\ 63 & 59 & 55 & 90 \\ 67 & 73 & 78 & 85 \end{bmatrix} \quad Q = \begin{bmatrix} 16 & 11 & 10 & 16 \\ 12 & 12 & 14 & 19 \\ 14 & 13 & 16 & 24 \\ 14 & 17 & 22 & 29 \end{bmatrix}$$

3. a) Compute the Golomb code for the following representation: (10) [CO3]
   i) $G_{12}(-15)$
   ii) $G^2_{exp}(13)$
   b) Apply pruning on the given binary image (Fig: 3(b)) using the structuring elements ($B1 \sim B8$) where- (15) [CO2]

$B1=$

x	0	0
1	1	0
x	0	0

$B5=$

1	0	0
0	1	0
0	0	0

$B2, B3$ & $B4$ are 90° rotated version of B_{i-1} from $B1$
$B6, B7$ & $B8$ are 90° rotated version of B_{i-1} from $B5$

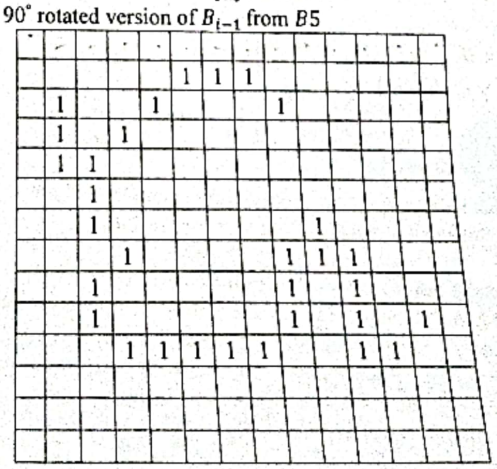


Fig: 3(b)

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c) Develop a mathematical model for degradation due to uniform linear motion. (10) [CO4]

4. a) Find the frequency domain filter $H(u,v)$ that works as a Laplacian filter in spatial domain, hence design a Wiener filter using $H(u,v)$ taking $SNR = 10$. (12) [CO3]

b) Explain the procedure of morphological reconstruction with its application. (10) [CO1]

c) A satellite captured an image of earth when one side of the earth is exposed to sunlight and other side remained dark, resulting a non-uniform illumination. Further, the memory of the satellite was lit by the cosmic rays causing an addition of very dense impulse noise to the stored image. What are the appropriate filters that can be used one after another to remove the impulse noises then to suppress the illumination? (13) [CO4]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Perform a full convolution on the following image using the given kernel. Be aware of the input and output indices. Use border replication padding while convolving. (10) [CO1]

	0	1	2	3
0	5	6	8	2
1	3	1	0	2
2	7	8	3	9
3	3	4	5	1

image

	-1	0	1
0	1	2	-2
1	-1	0	1

kernel

b) Prove that convolution with a 2D separable filter can be accomplished by performing two one-dimensional convolutions. (09) [CO2]

c) Suppose you want to segment the following image using normalized graph cut method proposed by Shi-Malik. Represent the image as a weighted graph considering 4-connected pixels and $\sigma_F = 4$; $\sigma_s = 1.5$, where symbols have their usual meaning. (16) [CO4]

8	4	9	25
5	11	22	28
9	2	27	30
29	27	28	31

6. a) Consider a grayscale image where foreground and background pixels are normally distributed with $\mu_f = 150, \mu_b = 50, \sigma_f = \sigma_b = 15$ and foreground pixels are one-third of total pixels. Find a minimum error threshold to separate the foreground from background. (09) [CO2]

b) For the following binary image, compute first two invariant *Hu* moments. Equations given below are for your convenience, where symbols bear usual meanings. (15) [CO3]

y \ x	0	1	2	3	4
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	1	1	0
3	1	1	1	1	0
4	0	0	1	1	0

$$\emptyset_1 = \eta_{20} + \eta_{02}; \emptyset_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2$$

$$\eta_{pq} = \frac{\mu^{pq}}{\mu_{00}^{\gamma}}; \gamma = \frac{p+q}{2} + 1$$

$$\mu_{pq} = \sum\sum (x - \bar{x})^p (y - \bar{y})^q f(x,y); \bar{x} = \frac{m_{10}}{m_{00}}; \bar{y} = \frac{m_{01}}{m_{00}}$$

$$m_{pq} = \sum\sum x^p y^q f(x,y)$$

c) Briefly explain Harris-Stephens corner detection method to find interest points from an image. Why is this method not covariant to image scaling? (11) [CO2]

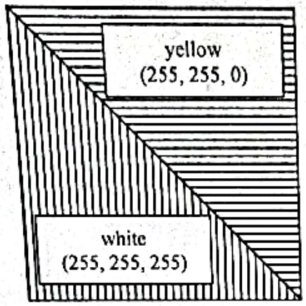
7. a) What does it mean by homography transform of an image? Explain how can you estimate homography matrix using direct linear transform from two images. (12) [CO2]

b) Following figure shows a RGB image with a white and a yellow region, where values in the parenthesis means RGB channel values. Compute edge magnitude and orientation at the diagonal line separating the regions using- (16) [CO3]

(i) Di Zenzo-Cumani color edge operator.

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(ii) First convert the image to grayscale then apply monochromatic edge operator. Use simplest possible derivative operator.



c) State the pros and cons of histogram based texture descriptors with proper example. (07) [CO3]

8. a) How can you modify the Lucas-Kanade optical flow estimation method to handle reasonably large motion? Explain with necessary illustration. Do you need image warping? If so, why? (13) [CO3]

b) The following table shows some features extracted from some binary images along with their known classes. If you would like to use them as training set of your Naïve Bayes classifier, then, estimate the classifier parameters. Also, determine the class of the test sample (0.5, 0.42, 260, 0.59). (16) [CO4]

Form factor	Roundness	Compactness	Eccentricity	Class
0.455	0.327	276	0.587	Star
0.469	0.416	268	0.26	
0.48	0.34	262	0.64	
0.453	0.29	277	0.75	
0.78	0.66	161	0.46	Hexagram
0.79	0.58	159	0.66	
0.82	0.54	153	0.67	
0.84	0.56	148	0.62	

[Hint: Estimate μ_{ji}, σ_{ji} for features x_j and class c_i]

c) Explain how histogram intersection can be used as a distance measure for feature descriptor matching. (06) [CO1]

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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) For the following two signals $f(x)$ and $g(x)$, find $f(x) \star g(x)$, where $\star$ is a convolution operator. (07) [CO1]

$$f(x) = \begin{cases} 1 & 0 \le x \le 1 \\ 0 & otherwise \end{cases} \qquad g(x) = \begin{cases} 1/2 & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$$

b) You are assigned to smooth the following image while preserving the edges. You planned to use a bilateral filter of size 3×3, and Gaussian function for both spatial kernel and range kernel. If you use $\sigma_s = 0.6$, and $\sigma_r = 15$ for those kernels, then show the spatial, range, and bilateral kernel for the encircled position of the following image. (16) [CO2]

$$\begin{bmatrix} 10 & 12 & 76 & 70 \\ 6 & \textcircled{5} & 66 & 68 \\ 4 & 25 & 55 & 62 \\ 8 & 9 & 71 & 58 \end{bmatrix}$$

c) For the given filter, determine the DTFT $H(u, v)$ of the filter (assuming the origin is at the center), and sketch the one-dimension profiles $H(u, 0)$ and $H(0, v)$. (12) [CO3]

$$\begin{bmatrix} -1 & -3 & -1 \\ 0 & 0 & 0 \\ 1 & 3 & 1 \end{bmatrix}$$

2. a) Show that Laplacian sharpening is a special case of an Un-Sharp Mask (USM) filter. (07) [CO3]
b) Use required mathematical formulation to show that the between class variance used by Otsu's thresholding method can be written as (14) [CO1]

$$\sigma_B^2(k) = \frac{(m_G P_1(k) - m(k))^2}{P_1(k)(1 - P_1(k))}$$

c) Estimate the optical flow for the following successive video frames using Lucas–Kanade method. Use masks suggested by Horn-Schunck to compute necessary partial derivatives. Also use a 2×2 spatial coherence window, and show the estimated result for encircled position only. (14) [CO2]

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 2 & \textcircled{9} & 9 & 8 \\ 1 & 9 & 9 & 10 \\ 0 & 9 & 8 & 9 \end{bmatrix}$$
Frame at t

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 2 & 10 & 9 \\ 1 & 1 & 9 & 8 \end{bmatrix}$$
Frame at $t + 1$

3. a) Geometric transformation of the following image is done by an inverse polynomial mapping function such as: $Y = WP(X)$, where, (13) [CO3]

$$X = [x_1 \quad x_2]^T, Y = [y_1 \quad y_2]^T,$$
$$P(x) = [1 \quad x_1 \quad x_2 \quad x_1^2 \quad x_1 x_2 \quad x_2^2], \text{ and}$$
$$W = \begin{bmatrix} 0.1 & 1 & 0 & 0.01 & -0.01 & 0.01 \\ 0.4 & 0 & 1 & 0.02 & -0.01 & 0.04 \end{bmatrix}$$

Determine output intensity at (1, 2), if bilinear interpolation is used during transformation.

x_2 \ x_1	0	1	2	3	4
0	100	105	108	109	102
1	99	104	109	100	103
2	200	204	205	102	105
3	201	106	101	103	111
4	206	208	204	200	100

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b) Let two interest points are described by some feature descriptors. Provide at least two matching ideas, with necessary equations, that can be used to decide whether those interest points are same or not. Also mention the pros and cons of those matching ideas. (12) [CO4]

c) Write Niblack's adaptive thresholding algorithm. Mention its use-cases. (10) [CO1]

4. a) During Hough transform based line detection, generally we get similar vote count in a small neighborhood. Explain what strategy can be applied to locate the peak voting position in the accumulator array. (06) [CO4]

b) Apply a 3×3 vector median filter on the location $(2, 2)$ of the given RGB image. (13) [CO2]

$$R = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 3 & 2 & 1 & 2 \\ 3 & 0 & 2 & 4 \\ 2 & 1 & 0 & 1 \end{bmatrix} \quad G = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 3 & 1 & 0 \\ 3 & 0 & 2 & 2 \\ 1 & 2 & 2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 3 & 1 & 2 \\ 2 & 3 & 0 & 1 \\ 1 & 0 & 2 & 1 \\ 3 & 1 & 2 & 2 \end{bmatrix}$$

c) Use DiZenzo-Cumani operator to get the edge strength and orientation for the following color image: values inside the parenthesis means (r,g,b) color vector. Also mention the importance of the operator compared to monochromatic edge detection technique for color image. Show result only for the center pixel. (16) [CO3]

$$\begin{bmatrix} (3,2,1) & (2,1,2) & (2,3,0) \\ (1,3,1) & (3,0,1) & (2,2,3) \\ (1,1,0) & (0,2,4) & (1,1,1) \end{bmatrix}$$

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

a) Define digital image processing. Briefly discuss the fundamental steps in digital image processing. (10) [CO1]

b) Consider the image segment shown below. (12) [CO2]

	3	1	2	1 ← q
	2	2	0	2
	1	2	1	1
p →	1	0	1	2

Compute length of the shortest-4, shortest-8 and shortest-m paths between pixels p and q where, $v = \{1, 2\}$. If a particular path does not exist between these two points, explain why.

c) Consider the following 3-bit image. (13) [CO3]

i) Generate the equalized image by using global histogram equalization processing.

ii) Plot the histogram of the processed image.

2	3	3	5	7
5	4	3	1	6
1	5	2	4	7
2	3	6	4	1
6	5	3	2	4

6. a) Define morphological gradient. Discuss when do we use top-hat and bottom-hat transformation in grayscale image. (10) [CO2]

b) Let, you are asked to enhance dark areas of an image while leaving the bright area as unchanged as possible. Propose strategy with necessary formulation to handle the problem. Use example to explain that the proposed method works. (14) [CO3]

c) Briefly discuss the following morphological image processing operations with necessary equations. (11) [CO3]

i) Convex hull.

ii) Pruning.

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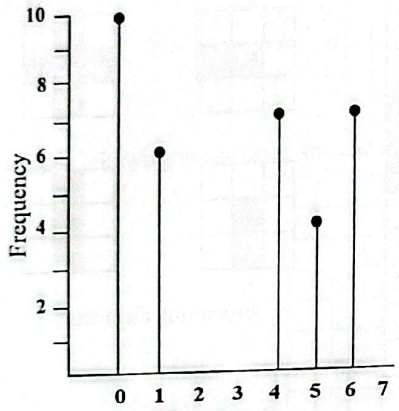
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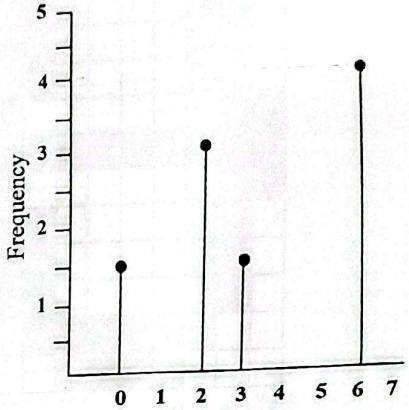
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Construct a modified image using histogram specification. The histograms of original and target images are given below. (20)



(a) Original Image



(a) Target Image

b) Use 8-connectivity and m-connectivity to find the shortest path between pixels (1, 1) with value 10 and (8, 8) with value 100, if V = {0, 10, 50, 65, 100, 106}. (10)

10	55	60	65	05
106	65	56	102	200
112	0	50	50	150
30	45	65	60	120
45	106	10	65	100

c) Let, you have computed the 2D DFT of a given image. The magnitude spectrum ranges from 0 to 10^5, but you want to display the image as an 8-bit gray scale image. For this purpose, you apply base-10 log transformation to the spectrum. What must be the normalizing constant of the transformation? (05)

2. a) Following is a 5×5 image with 5 distinct gray levels: (25)

18	18	4	12	12
10	10	4	12	4
10	4	12	4	15
12	10	15	15	10
15	12	4	10	4

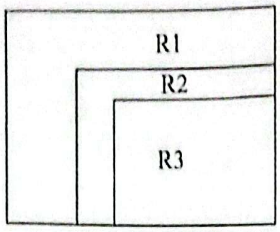
Compare the efficiency of compressing the image using Huffman encoding and Golumb coding with parameter, $m = 5$.

b) Explain why transformation function to perform image complement approximation is different for each component in HSI model but similar for all components in RGB model. (10)

3. a) Consider a line of intensity values, {100, 129, 145, 56, 99} If it is quantized with 4 higher bit planes, compute the error and signal-to-noise ratios for the quantized data. (08)

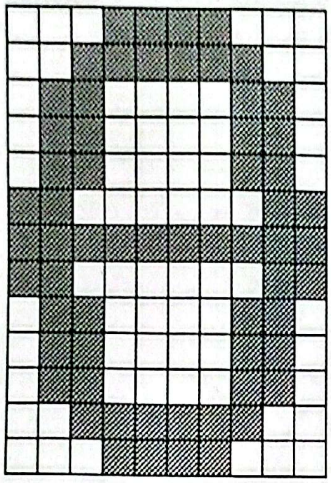
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b) Explain which type of histogram equalization would perform better visual improvement for the following image. (12)

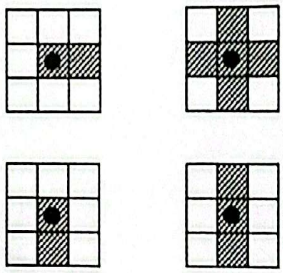


Pixel values of region, R1 = 127, R2 = 129, R3 = 130.

c) Figure below shows a binary image containing Greek letter θ, and four structuring elements. What morphological operations(s) would you use to obtain the number zero (0) from the letter θ? Consider that you can use any of the given structuring elements as many times as you wish. Provide a reason for your choice of operation(s) as well as the structuring elements. (15)

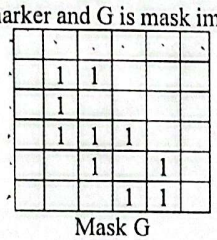


Binary image

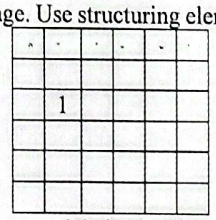


Structuring elements

4. a) Sketch the result of the following operations: (10)
$D_G^{(n)}(F)$ where F is marker and G is mask image. Use structuring element of 3×3 with all 1's.



Mask G



Marker F

b) Use necessary diagrams to show the conceptual relationship between the RGB and HSI color models. (12)

c) Explain with necessary mathematical formulation, how you can apply a linear filter on a color image represented in HSV color space. Also mention the need of saturation weighted hue component. (13)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) State the property of a linear operation. With necessary mathematical formulation, show that convolution is a linear operation. (10)

b) You are given the following spatial domain filter. (15)

0.15	0.35	0	−0.35	−0.15

i) Compute Fourier spectrum and phase of this filter,

ii) What do you think the effect of applying this filter to an image will be? Justify your answer.

c) Construct a homomorphic filtering function using a Butterworth high pass filter to correct nonuniform illumination. (10)

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6. a) Consider the image histogram given below. (18)

Intensity level	0	1	2	3	4	5
Frequency	14	11	4	12	16	7

Use maximum entropy-based thresholding technique to find a global threshold to segment the image.

b) State the necessity of non-maximum suppression stage used in Canny edge detection algorithm. Provide an idea to get edge orientation by avoiding trigonometric function that can be used in the Canny's algorithm. (12)

c) "Median filter may reduce the number of occupied bins in a histogram." – justify the statement. (05)

7. a) A 100×100 image is warped applying the following affine transformation matrix T. Mention the warped image size. Let, you are currently at pixel (50, 65) in the warped image, find its intensity if inverse mapping technique and bilinear interpolation were used during warping. Consider the following image segment of the input image. (16)

$$T = \begin{bmatrix} 1 & 0.6 & 5 \\ 1.2 & 0.8 & 10 \\ 0 & 0 & 1 \end{bmatrix}$$

y \ x	36	37	38	39
10	60	65	70	72
11	66	70	80	88
12	80	90	95	120
13	90	92	110	150

image segment

b) Segment the following image using spit and merged procedure. (12)

	1	1	1	1	1	
	1	1	1	1	1	
	1			1	1	
	1			1	1	

c) Explain aperture problem in optical flow estimation. (07)

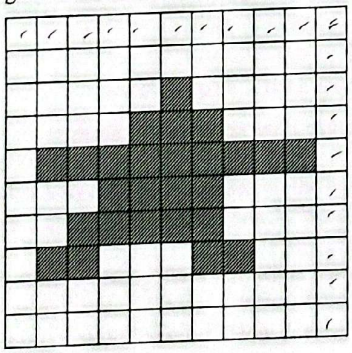
8. a) Let you are going to restore a noisy image having only additive noise. For the restoration purpose you construct the following histogram of a flat region of the image. (12)

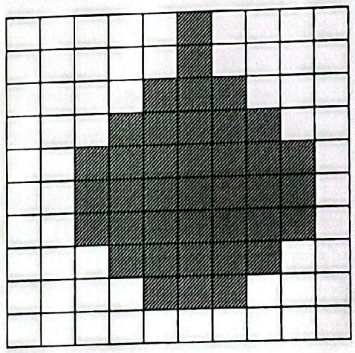
Intensity level	0	1	2	3	4	5	6	7
Probability	0	0	0.210	0.404	0.277	0.102	0.022	0.003

i) Determine the type of noise present,

ii) Provide an estimation of the parameters: mean and standard deviation of the noise density function.

b) Find the region descriptors: form factor, compactness, and circularity to describe the binary region in the following images. (12)





c) Derive the brightness consistency equation to estimate motion field as a form of optical flow. Suggest an idea to overcome its limitation. (11)

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SECTION A

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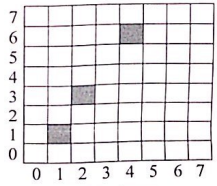
1. a) Derive the equation for Laplacian of a 2D Gaussian function $G = A \cdot e^{-\frac{x^2+y^2}{2\sigma^2}}$. Sample the LoG function to get a convolution kernel with $A = \sigma^2 = 1$. Scale the kernel values to the nearest integer. Use appropriate kernel size to support the function. (17)

b) Determine the 4-point 1D DFT matrix. Use it to find the 2D DFT of the following image. (13)

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & -1 \\ -1 & 2 & 1 & 0 \\ 2 & -1 & 1 & 1 \end{bmatrix}$$

c) Explain an implementation idea for hysteresis thresholding used in Canny edge detection method. (05)

2. a) Use Hough line transformation for the following edge image to detect the line. Use an appropriate accumulator array with $d\theta = 30°$, $d\rho = 1$. θ spans from 0° to 180°, ρ spans from -4 to 8. Show vote count in the array and determine the line equation. (18)



Edge Image

b) Use weighted median filter to restore the following 4-bit image. Show results only for the middle row, use the given weight matrix. (10)

4	3	1	2	6
2	7	3	5	0
9	15	6	0	3
12	4	10	8	7
11	6	2	7	6

Image

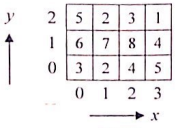
1	2	1
2	3	2
1	2	1

Weight Matrix

c) If $x_0(t) = at/T$ and $y_0(t) = bt/T$ denotes the time varying components of motion in x and y directions, then show that uniform linear motion between image and the sensor during image acquisition can be modeled by the following degradation function. Here, symbols have their usual meanings. (07)

$$H(u,v) = \frac{T}{\pi(ua+vb)} \sin[\pi(ua+vb)] e^{-j\pi(ua+vb)}$$

3. a) What is the importance of image warping in computer vision? Warp the following image with the given transformation matrix, T. Use backward mapping and bilinear interpolation during the process. Mention the size of the warped image and determine the intensity only at pixel (2, 3). (16)



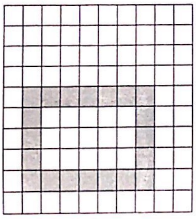
Image

$$T = \begin{bmatrix} 0.43 & -0.25 \\ 1 & 1.73 \end{bmatrix}$$

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b) State the maximum entropy based image thresholding algorithm with necessary mathematical derivation. (16)

c) Draw signature for the following binary image. (03)



Binary Image

4. a) State the pros and cons of histogram based texture descriptors with proper example. (08)

b) For the following image, determine the Gray Level Co-occurrence Matrix (GLCM) with displacement $d = [1, 1]$. And thereby, measure the uniformity and homogeneity of the matrix. (17)

y \ x	0	1	2	4	1
1	4	0	2	3	
1	1	3	2	0	
3	2	0	3	1	
4	3	2	4	2	

c) Derive the equations used by Lucas-Kanade to estimate motion field from a video sequence. (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Illustrate the 'Golomb Coding' technique to encode non-negative integer in data compression. Calculate the Golomb Codes for the following representation: (20)

i) $G_{11}(37)$

ii) $G_{12}(-15)$

iii) $G^2_{exp}(13)$

b) Explain false contouring effect in digital image using the concept of sampling and quantization. (08)

c) What measure can be taken to improve the brightness discrimination provided that a weber ratio is given? (07)

6. a) Explain the application of pseudo-color processing for highlighting explosives contained in a luggage using proper transformation functions. (12)

b) Write down the conditions for pre-color-component processing and vector-based processing to be equivalent with proper example. (08)

c) Derive the CMY intensity mapping function $s'_i = kr'_i + (1 - k)$ using its RGB counterpart where $0 < k < 1$. (10)

d) Prove, erosion and dilation are duals of each other with respect to set complementation and reflection. (05)

7. a) Write down the technique for filtering the hue component in HSI color space provided that the saturation is low and hue value is quite unstable. (12)

b) 'It is possible to save a digital image with roughly 50% less storage than the original image using bit plane slicing.' – Justify the statement. (08)

c) Illustrate the objective fidelity criteria for quantifying the nature of loss in image compression. (10)

d) Mention the drawbacks of 'Convex Hull' algorithm. (05)

8. a) Illustrate the automated hole filling algorithm for binary image based on morphological reconstruction. (12)

b) Suppose, you are asked to enhance dark areas of an image that may contain hidden features while leaving the light area as unchanged as possible. Now, apply 'local enhancement by contrast manipulation' to solve this problem using histogram statistics. (15)

c) If histogram equalization is applied twice to an image, that is, if it is applied to the result of histogram equalization, will the second application change the image or not? Explain why or why not? (08)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2020
Department of Computer Science and Engineering
CSE 4127
Image Processing & Computer Vision

TIME: 1.5 hours FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) Consider the following two gray scale images (Image *A*, Image *B*) of size $N \times N$. Although they look quite different, their histograms are identical. If each of the images are blurred with the Mask *M*, then (10)
   i) Would the resultant histograms still be the same? Explain.
   ii) Calculate the approximate histograms of the resultant images.

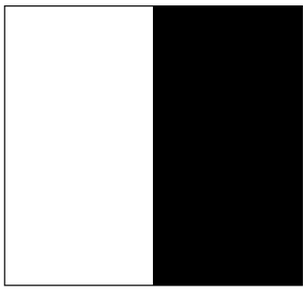


Image: *A*

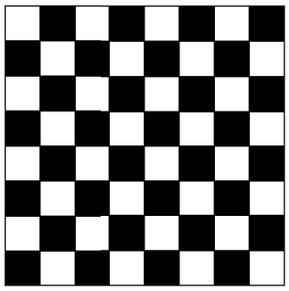


Image: *B*

$$\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$$

Mask: *M*

b) Let an image *A* has gone through an intensity transformation function $S = T(r) = cr^{\gamma}$, where c, γ are constants to get the output image *B*. Some pixel intensities of *A* and *B* are shown below. (13)
   i) Find the missing input intensities P, Q.

	4		15
		Q	
P		16	

$\longrightarrow T(r) \longrightarrow$

	12		87
		105	
22		96	

Image: *A* Image: *B*

c) Do you think the discrete histogram equalization technique is generally able to produce a flat histogram? Justify your answer. (07)

2. a) Suppose an image with intensities in the range [0,1] has the PDF $P_r(r)$ as shown in Fig. i. Assume continuous quantities and find the transformation in terms of r and z that will transform the intensity levels of the image so that they will have the specified $P_z(z)$ as shown in Fig ii. (12)

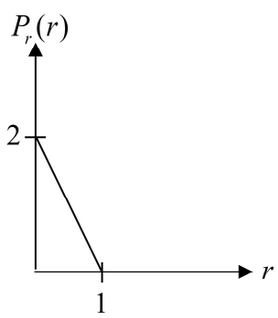


Fig. i

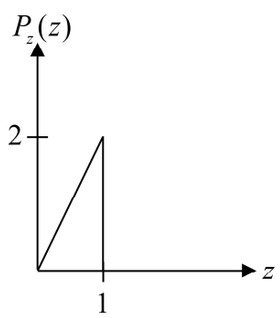


Fig. ii

b) Explain convex deficiency. Is it possible to reduce the hit-or-miss transform to simple erosion? Justify your answer. (12)

c) What does it mean by sampling and quantization in digital image acquisition? (06)

3. a) Consider the following 4×8 image. (15)

21	169	169	243	21	95	132	95
243	169	132	95	109	169	109	132
95	21	132	169	169	95	109	109
95	132	169	21	95	243	132	169

i) Compress the image using Huffman coding.

ii) Calculate the compression ratio and the percentage of the original fixed-length representation that is removed as the coding redundancy.

b) Calculate the morphological thinning on image E using P as the initial structuring element and show the outcome of each step separately. (15)

0	0	0	0	0	0	1	1
1	0	0	0	0	0	1	1
1	1	1	1	1	0	0	0
1	1	1	1	0	1	1	0
1	1	1	1	1	1	0	0
1	1	1	0	1	1	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Image: E

0	0	0
×	1	×
1	1	1

Structuring Element: P

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Let a Gaussian function $G(x,\sigma)=e^{-x^2/2\sigma^2}$ and its derivative G' is sampled at 5 points symmetric to the origin with $\triangle x=1,\sigma=1$. The sampled values of G and G' are used as $1D$ separable kernel to convolve an image in x and y direction, respectively. Find (14)

i) The x and y directional $1D$ kernels; scale and round the values to integer.

ii) The $2D$ kernel from $1D$ kernels.

iii) Note the benefits of using separable $2D$ kernels.

b) Consider the following 4-bit image corrupted with impulse noise. To remove the noise (16)

i) Use a contraharmonic mean filter of size 3×3 with appropriate order Q.

ii) Use a suitable order-statistics filter of size 3×3.

Show results only for the middle row.

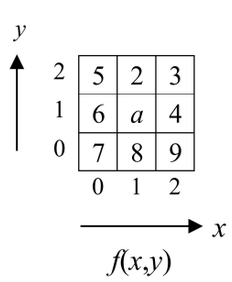
7	6	8	5	4
3	10	9	8	11
8	x	11	12	1
5	8	2	10	7
9	3	4	15	14

Here, $x=\begin{cases}0; & \text{if your roll is an odd number}\\ 15; & \text{Otherwise}\end{cases}$

5. a) Let you are assigned to warp the following image $f(x,y)$ with the transformation matrix T. You are also constrained with the conditions that the transformation should be done using backward mapping technique having bilinear interpolation and the warped image should be large enough to contain all the processed samples. If so (14)

i) Find the size of the warped image.

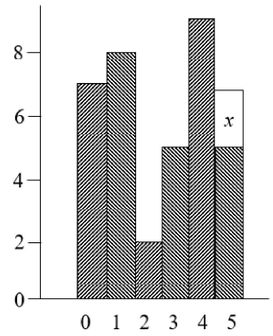
ii) Find the intensity at pixel (1,3) of the warped image.



$$T=\begin{bmatrix}1 & -0.5\\ 0.5 & 1\end{bmatrix}$$

Here, $a=$ last digit of your roll + 1.

b) An image intensity distribution shown by the following bar chart. Find a global optimal threshold for the image using Otsu's method. (16)



Here, $x =$ sum of last two digits of your roll number.
Therefore, $frequency(5) = x + 5$.

6. a) Let you have planned to describe an image texture as an Entropy of Gray Level Co-occurrence Matrix (GLCM). For the given image below (15)
   i) Find the GLCM with displacement $d = [1,0]$.
   ii) Determine the measure of Entropy.

0	1	4	2	2
3	4	3	0	1
1	1	4	2	0
3	2	4	3	1
4	3	0	2	2

b) Using a 2×2 window, estimate the optical flow using Lucas-Kanade method for the following two successive frames of a video sequence. Show result for (1,1) position only. (15)

	0	1	2	3	→ x
0	1	1	0	1	
1	1	9	9	9	
2	1	9	10	9	
3	0	9	9	10	
↓ y					

Frame at t

	0	1	2	3	→ x
0	1	1	0	1	
1	0	a	1	0	
2	1	1	9	9	
3	0	1	9	9	
↓ y					

Frame at $t+1$

Here, $a = \begin{cases} 0; & \text{if your roll is even} \\ 1; & \text{Otherwise} \end{cases}$

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Department of Computer Science and Engineering
CSE 4127
Image Processing and Computer Vision

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Specify image compression. 'Huffman Coding can be used as a symbol encoder in an image compression system'-illustrate the statement. (12)
   b) Explain false contouring with necessary examples using the concept of sampling and quantization. (08)
   c) Consider the image segment shown below: (10)

4	2	3	2 (q)
3	3	1	3
2	3	2	2
(p) 2	1	2	3

Complete length of the shortest-4, shortest-8, and shortest-m paths between pixels p & q where, v = {1, 2}. If a particular path does not exist between these two points, explain why?
   d) Interpret the significance of shading correction in digital image processing. (05)

2. a) Construct a modified image using histogram specification where histogram of original and target images are given below: (20)
original histogram:

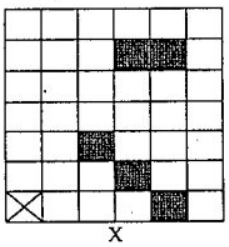
Gray level	0	1	2	3	4	5	6	7
No. of Pixels	8	10	10	2	12	16	4	2

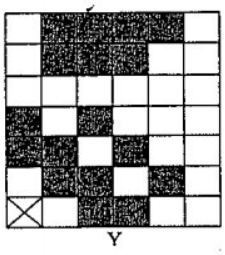
Target histogram:

Gray level	0	1	2	3	4	5	6	7
No. of Pixels	0	0	0	0	20	20	16	8

   b) Assuming continuous intensity values, suppose that an image has the intensity PDF $P_r(r) = {}^{2r}/_{(L-1)^2}$ for $0 \le r \le (L-1)$ and $P_r(r) = 0$ for other values of r. find the transformation function that will produce an image whose intensity PDF is $P_z(z) = {}^{3z^2}/_{(L-1)^3}$ for $0 \le z \le (L-1)$ and $P_z(z) = 0$ for other values of z. (10)
   c) 'Local Histogram processing is necessary to enhance details over small areas in an image'-justify the statement. (05)

3. a) Image X and Y are shown below with crossing in the left bottom showing their correspondence. (12)


X


Y

Design an appropriate structure element such that Y can be obtained after dilating X. (Black for 1 and white for 0).
   b) 'Textural segmentation is crucial in finding a boundary between two regions based on textural content'-justify the statement with necessary figures. (11)
   c) Detect edge on the following image using erosion and show the outcome of each step separately. (12)

0	0	1	1	1	1	0	1	1	1	0
0	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	0
1	1	1	1	0	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	0
0	0	0	0	0	1	1	1	0	0	0

0	1	0
1	1	1
0	1	0

Structuring Element

4. a) State and explain the data redundancies that can be identified in an image for the exploitation of image compression. (09)
   b) Define Hue and Saturation. Calculate the HSI value of the following RGB point: (11)
   (155, 210, 75)
   c) What will happen if you change the value of H channel of an HSI image while keeping the other channel's value constant? (05)
   d) Illustrate the concepts: (i) Color slicing and (ii)Tone correction (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Compare between image processing and computer vision. Explain some typical challenges in solving a computer vision problem. (12)
   b) Consider the 1D image below: (08)

a	b	c	d	e	f	g

   i) Suppose, you apply a width-3 mean filter twice. Give an expression for the center pixel after the filter has been applied.
   ii) Define a single 1D kernel that, when applied only once to the image, will produce the same results.
   c) Find the unsharp mask first and then the sharped image for the following image: (15)

2	8	6
5	4	1
7	3	9

   Use a 3x5 mean filter as a blurring kernel and do wrap around padding where required.

6. a) Design a Bulterworth Low-pass Filter (BLPF). What happens when you choose a higher order BLPF to smooth an image? When would you prefer BLPF than Gaussian Low-pass Filter? (12)
   b) Show that the Fourier transform of a shifted function is unaltered except for a linearly varying phase factor. (08)
   c) Verify the soundness of the statement: 'Laplacian with a -8 in the center yields sharper results than the one with -4 in the center'. (10)
   d) Explain estimation of image degradation function by mathematical modeling. (05)

7. a) Segment the image shown below with a global threshold computed by Otsu's method. (13)

1	2	0	12	14
2	3	4	8	13
9	2	3	15	7
3	2	2	11	13
1	2	0	14	12

   Mention some limitations of global thresholding.
   b) Rotate the image shown below by backward mapping method with $\theta = 30^{\circ}$. (12)

1	5	2
6	3	4
7	9	5

   Use bilinear interpolation and allow the image to grow to fully contain the rotated image.
   c) Find the 4-point basis matrix of Discrete Fourier Transform (DFT) and then use it to find the DFT of [1 -1 1 -1]. (10)

8. a) Define image representation and description. Measure the entropy and uniformity of the following 3-bit image: (13)

1	2	0	7	5
6	2	3	4	3
4	0	2	1	5
4	7	6	2	4
5	6	4	3	3
5	4	1	2	4

   b) Explain the basic principle of Hough line detection mechanism. (09)
   c) What is optical flow? Explain Lucas-Kanade method of calculating optical flow. (13)

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CSE 4127
Image Processing and Computer Vision

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

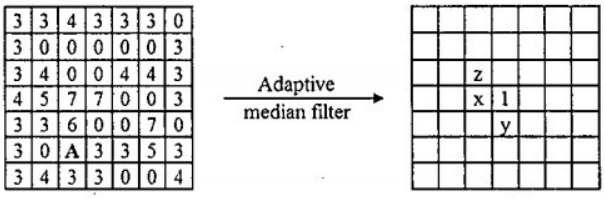
(Answer **ANY THREE** questions from this section in Script A)

1. a) Compare and contrast between image processing and computer vision. Mention some applications of them. (08)
   b) Use Laplacian operator to sharp the following image. Use necessary border reflection padding. (15)

$$\begin{bmatrix} 1 & 2 & 5 & 4 \\ 5 & 2 & 5 & 5 \\ 2 & 1 & 6 & 3 \\ 2 & 4 & 7 & 6 \end{bmatrix}$$

   c) What is periodic noise? Design a band reject and a notch reject filter to remove such noise from image. Also discuss their pros and cons. (12)

2. a) Given an input image of size 7x7 shown below, was filtered using 3x3 adaptive median filter with maximum allowed size of 5x5. (12)



input image output image

Answer the following questions with reasoning:
i. What is the value of the pixel A in the input image?
ii. What are the values of pixels x, y and z in the output image?

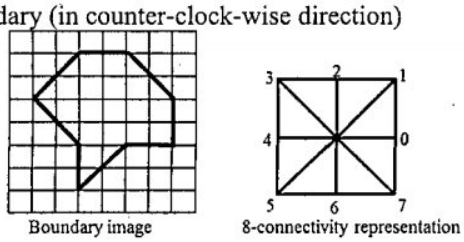
   b) Explain why and how non-maximum suppression is performed in Canny edge detection algorithm. (10)
   c) Let a 3-bit grayscale image is given below: (13)

0	5	2
6	1	7
3	4	7

Resize the image by using bilinear interpolation and backward mapping to obtain 3x5 image.

3. a) Show that $\nabla^2 G = \frac{1}{\pi\sigma^4}\left[1-\frac{x^2+y^2}{2\sigma^2}\right].e^{-\frac{x^2+y^2}{2\sigma^2}}$, where G is the Gaussian function and ∇^2 is the Laplacian operation. (08)
   b) What is Thresholding? List the steps of Otsu's Thresholding algorithm. (11)
   c) Compare the first order and second order derivative based edge detection methods. (10)
   d) State and prove the separability property of 2D DFT. (06)

4. a) For the following boundary (in counter-clock-wise direction) (09)



Boundary image 8-connectivity representation

compute the 8-connected
i. chain code.
ii. chain code that is invariant to rotation.
iii. chain code that is invariant to both starting point and rotation.

b) Consider two consecutive frames of a video shown below. (14)

y \ x	0	1	2	3	4
0	3	15	6	4	0
1	4	4	12	1	2
2	4	14	5	5	8
3	6	1	13	2	3
4	10	7	11	9	12

Frame - 1

y \ x	0	1	2	3	4
0	4	6	5	3	0
1	2	3	15	6	4
2	3	4	5	12	1
3	10	4	13	5	5
4	9	6	1	13	2

Frame-2

Estimate the optical flow of the center pixel using Lucas-Kanade method. Use 3x3 window for spatial coherence.

c) Explain inverse filtering process to restore a noisy image. What is the shortcoming of inverse filter? How does Wiener filter overcome such shortcoming? (12)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is digital image and digital image processing? What are the fundamental steps in digital image processing? (10)

b) Define blind spot and brightness adaption. Briefly describe how to characterize brightness discrimination using appropriate example. (08)

c) What do you mean by image compression? Briefly describe about Discrete Cosine Transformation of image compression. (12)

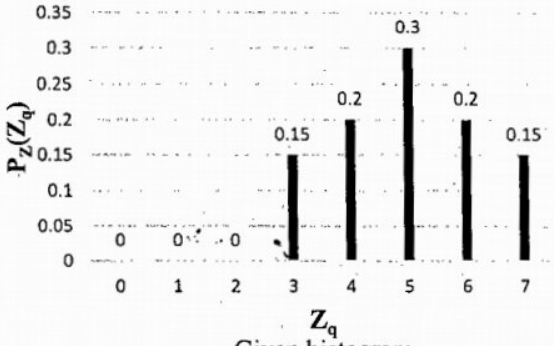
d) What are the changes occur in an image by changing values in spatial domain and quantization (Gray value)? (05)

6. a) Define contrast. Design a mapping function to enhance the contrast of an image having very bright pixels. (11)

b) Briefly describe Power-law transformation for digital image processing and also show the image appearance of different values of gamma. (08)

c) The following table shows the information of a 3-bit 64x64 image. (16)

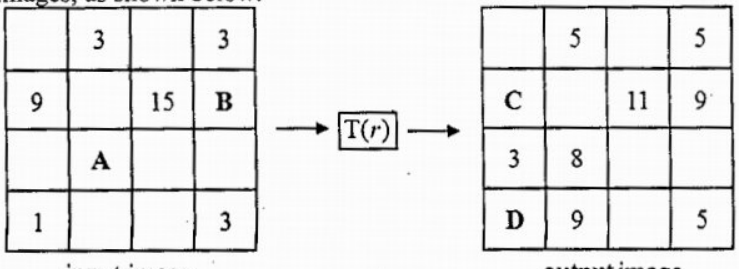
Pixel Value	Frequency
0	790
1	1023
2	850
3	656
4	329
5	245
6	122
7	81



Given histogram

Draw the original histogram, normalized histogram and finally show the matching histogram with the given histogram.

7. a) Let a 4-bit, 4x4 image passes through an intensity transformation function given by $s = T(r) = \alpha \log_2(1+r) + \beta$, where α, β are constants. Few pixels are available in the input and output images, as shown below. (10)



input image output image

What are the values of the pixels A, B and C, D in the input and output images respectively?

b) Define Pseudo color image processing. Discuss about intensity slicing with appropriate diagram. (10)

c) Write down the procedure to compute pruning of a binary image. (05)

d) Consider the following 5x5 8-bit image with grayscale range of [0, 9]. (10)

$$\begin{bmatrix} 0 & 5 & 7 & 7 & 5 \\ 7 & 2 & 6 & 2 & 6 \\ 6 & 9 & 7 & 7 & 0 \\ 6 & 6 & 1 & 7 & 6 \\ 9 & 3 & 2 & 4 & 8 \end{bmatrix}$$

Compress the image using Huffman encoding techniques and also discuss the efficiency of this compression.

8. a) Consider the following binary image A. (12)

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Image A

Show the result of applying the following operations on image A with a 3x3 structuring elements.

i. $B = (A \oplus S) \cap A^C$

ii. $B = (A \ominus S) \cup A^C$

iii. $B = (A \oplus S) \cup A^C$

b) How can you detect how many template images are there in a full image using hit or miss transformation algorithm? Briefly describe with appropriate example. (11)

c) Define Hue and Saturation. Write down the process of conversion of a pixel value from HSI color space to RGB color space. (12)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2017
Department of Computer Science and Engineering
CSE 4127
Computer Vision and Image Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Relate and differentiate between computer vision and image processing. Explain the possible challenges one has to face in solving a computer vision problem. (10)
   b) Define convolution. Show by using an example that $f \star e = f$, where f is an image, e is a unit impulse, and $\star$ is a convolution operator. (10)
   c) Sharp the following image using unsharp mask. Use a 3×3 Gaussian blurring kernel with $\sigma = 1$ and do zero padding where necessary. (15)

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 5 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix}$$

2. a) What is the necessity of transforming an image into frequency domain? Let, $F(u, v)$ is a [M, N] point Discrete Fourier Transform (DFT) of an image defined by $f(x, y)$. Then show that $F(u, v) = F(u + pM, v + qN)$, where p, q are arbitrary constant. (10)
   b) Describe Canny edge detection algorithm in detail. (12)
   c) Design a Butterworth Lowpass Filter (BLPF) to smooth an image. How can you choose the cut-off frequency D_0 to pass α percent of image power? Also explain why the lower order BLPF is more preferable than higher order one. (13)

3. a) Define order statistic filter. Use a 3×3 alpha-trimmed mean filter with $d = 4$ to filter noises from the image presented below. (13)

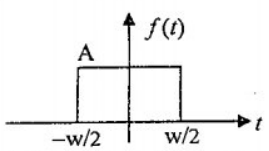
$$\begin{bmatrix} 2 & 3 & 5 & 8 & 7 \\ 8 & 12 & 4 & 6 & 3 \\ 5 & 7 & 6 & 3 & 9 \\ 4 & 1 & 4 & 8 & 3 \\ 6 & 7 & 8 & 14 & 17 \end{bmatrix}$$

   b) What is periodic noise? How can you remove such noise from an image? (05)
   c) Explain inverse filtering process to restore a noisy image. What is the shortcoming of inverse filter? How does Wiener filter overcome such shortcoming? (12)
   d) What is image segmentation? Mention some applications of it. (05)

4. a) Find a global threshold for the following image. What are the problems of using a single global threshold to segment an image? How can these be resolved? (13)

$$\begin{bmatrix} 1 & 2 & 1 & 0 & 12 & 14 \\ 2 & 3 & 4 & 8 & 13 & 12 \\ 9 & 2 & 3 & 15 & 13 & 7 \\ 3 & 2 & 2 & 10 & 11 & 13 \\ 1 & 2 & 0 & 5 & 12 & 14 \\ 2 & 3 & 6 & 8 & 15 & 13 \end{bmatrix}$$

   b) Perform Fourier transform of the function shown in the following figure. The transformation equation is given by $F(u) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi ut} dt$. (08)



   c) Define motion field and optical flow. Explain the Lucas-Kanade optical flow estimation method stating their key assumptions. (14)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is a digital image? What are the stages through which an image passes in a image processing system? Briefly explain each stage. (10)
   b) Briefly discuss about Weber law and optical illusions. (09)
   c) What is mean by sampling and quantization? Write down the transformation matrices of image Scaling, Rotation and Translation. (08)
   d) Define image compression. Briefly describe any one of the image compression techniques. (08)

6. a) Define bit plane slicing for piecewise linear transformation and histogram equalization. Calculate each plane for the following 4-bit image. (10)

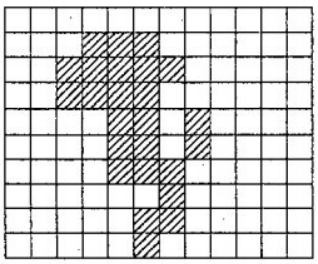
$$\begin{bmatrix} 11 & 13 & 3 & 5 & 8 \\ 8 & 6 & 0 & 11 & 10 \\ 9 & 0 & 2 & 9 & 7 \\ 7 & 12 & 6 & 13 & 14 \\ 3 & 4 & 10 & 0 & 2 \\ 8 & 0 & 15 & 5 & 1 \end{bmatrix}$$

   b) Consider the following 5×5 image with grayscale range of [0, 9]. (17)

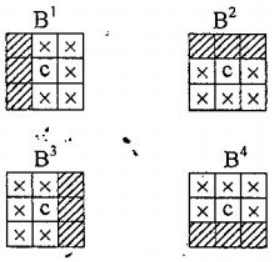
$$\begin{bmatrix} 0 & 5 & 7 & 7 & 5 \\ 7 & 2 & 6 & 2 & 6 \\ 6 & 9 & 7 & 7 & 0 \\ 6 & 6 & 1 & 7 & 6 \\ 9 & 3 & 2 & 4 & 8 \end{bmatrix}$$

      i) Plot the histogram of the image.
      ii) Perform histogram equalization and then show the equalized image and plot histogram again.
      iii) Histogram matching is a useful contrast manipulation technique that transforms an image's histogram to match with that of another image. Clearly describe how can you achieve it.
   c) What do you mean by contrast stretching? Explain a simple non-linear contrast stretching method. (08)

7. a) Define full color image processing. Discuss about intensity slicing with appropriate diagram. (08)
   b) Define convex hull. Draw the convex hull of the following binary image. (17)



Binary imge: A



Structure elements: c is the center

   c) Define morphological gradient. When do we use top-hat and bottom-hat transformation in grayscale image? (10)

8. a) Define pigments. Write down the process of conversion of pixel value from RGB color space to HSI color space. (16)
   b) Prove that in continuous domain, probability distribution function of equalized image is always uniform. Discuss it with appropriate example. (07)
   c) Consider a portion of a binary image as shown below. Perform the following operations on the image. (12)
      i) Opening
      ii) Closing

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Binary image

[① 1 1]

Structuring Element (SE)

The circle indicates the center of SE

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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is computer vision? How is image processing related to computer vision? Mention some challenges of computer vision problem. (10)

b) Define histogram equalization. Consider the 5×5, 3 bits/pixel image shown in Fig. 1(b). (18)

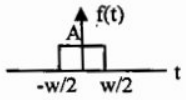
$$\begin{bmatrix} 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \\ 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 4 & 5 & 3 \\ 4 & 3 & 2 & 6 & 7 \end{bmatrix}$$

Fig. 1(b)

i) Perform histogram equalization to the image.
ii) What will happen if we apply the same process on the resultant image of step (i).
iii) Explain why the discrete histogram equalization does not, in general, yield a flat histogram.

c) Perform Fourier transform of the function shown in the following figure follows from (07)

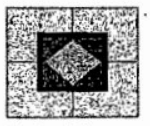
$$F(\mu) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi\mu t}dt$$



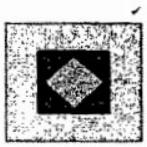
2. a) Prove that the convolution of an image in spatial domain is equal to multiplication in frequency domain. (13)

b) What does it mean by low pass filter? Design an Ideal Low-Pass Filter (ILPF). Explain why there is a ringing effect on the image that is smoothed by ILPF. (10)

c) Consider the following figure, when the image on the left was filtered using a smothing filter, the result was image on the right. The filter used was one of these of size 3×3: (i) mean filter; (ii) ideal low-pass filter; (iii) Gaussian low-pass filter; (iv) median filter. For each of the four possible filters listed above, give at least one reason why you think it was or was not, the filter actually used. (12)



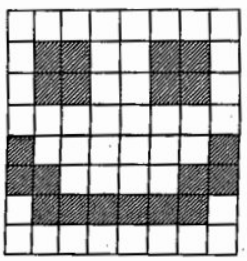
original image



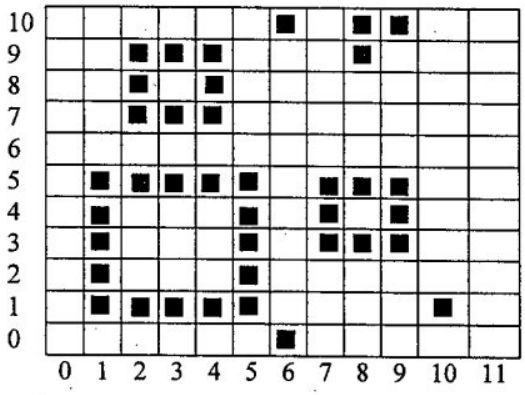
filtered image

3. a) Sketch and explain a model for image restoration process. How can you estimate the image degradation functions? (12)

b) What is the importance of image segmentation? Segment the binary image shown in following figure. Using split-and-merge technique. Also, show the quad tree corresponding to the segmentation. (12)



c) If we apply the Hough transform on the following image, what would be the maximum value for the accumulator cell in the (P,Q) space? What is the corresponding (P,Q) value? (11)



4. a) Explain how the k-means clustering can be used for image segmentation. What are the weaknesses of k-means clustering algorithm? How can they be overcome? (13)
   b) What is motion field? Explain a direct method for estimation of motion field stating the key assumptions and deriving the necessary equations. (15)
   c) State the steps of calculating Ostu's threshold for an image. (07)

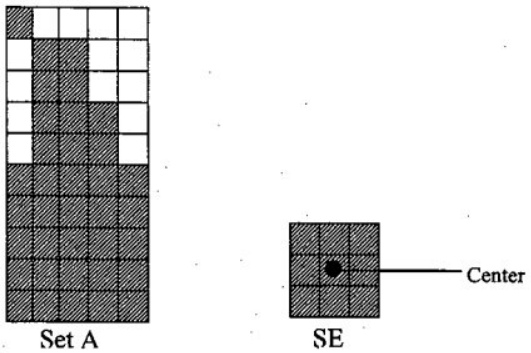
SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is digital image processing? What are the fundamental steps in digital image processing? (09)
   b) Discuss about different neighborhood and connectivity of a pixel with appropriate example. (12)
   c) Define blind spot and brightness adaption. State some example of optical illusions. (06)
   d) What do you mean by image compression? Briefly describe any one of the image compression techniques. (08)

6. a) What is discretization? How are images presented in computer? (07)
   b) What is the property of an adaptive filter? Write a pseudocode for adaptive median filter. Instead of median filter, when will you prefer the adaptive one? Explain why. (12)
   c) State and explain the data redundancies that can be identified in an image and can be exploited for image compression. (10)
   d) Explain how human visual system perceives a color. (06)

7. a) Define convex hull. Write down the drawbacks of convex hull and how this can be resolved? (07)
   b) Define Thinning. Draw the skeleton of the following binary image. Is it possible to recover the original image from the skeleton? (15)



   c) Why pruning is necessary? Write down the equations to compute pruning of a binary image. (06)
   d) Define Erosion and Dilation for Gray-scale image with appropriate example. (07)

8. a) When we use top-hat and bottom-hat transformation in grayscale image? (04)
   b) Define Hue, saturation and intensity. Write down the process of conversion of a pixel value from RGB space to HSI space. (18)
   c) Define pseudo color image processing. Discuss about intensity slicing with appropriate diagram. (08)
   d) How can you apply a spatial filter on a full color image? Explain. (05)